

Unitary Matrix

Unitary Matrix is a square matrix of complex numbers. The product of the conjugate transpose of a unitary matrix, with the unitary matrix, give

Let us learn more about the properties, and examples of the unitary matrix.

What is Unitary Matrix?

A **unitary matrix** is a square matrix of complex numbers, whose inverse is equal to its conjugate transpose. Alternatively, the product of the uni

- $U^H = U^{-1}$
- $U^H U = U U^H = I$

where I is the identity matrix whose order is the same as U .

Also, a unitary matrix is a nonsingular matrix. Or the determinant of a unitary matrix is not equal to zero. The columns and rows of a unitary mat

Properties of Unitary Matrix

The properties of a unitary matrix are as follows.

- The unitary matrix is a **non-singular matrix**.
- The unitary matrix is an invertible matrix
- The product of two unitary matrices is a unitary matrix.
- The inverse of a unitary matrix is another unitary matrix.
- A matrix is unitary, if and only if its transpose is unitary.
- A matrix is unitary if its rows are orthonormal, and the columns are orthonormal.
- The unitary matrices can also be non-square matrices but have orthonormal columns and rows.

Note: The sum or difference of two unitary matrices doesn't need to be a unitary matrix. For example, if A is a unitary matrix, then $A - A = O$

Terms Related to Unitary Matrix

The following terms related to matrices are helpful for a better understanding of this concept of unitary matrix.

- **Non-Singular Matrix:** The **determinant** of a non-singular matrix is a non-zero value. For a square matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the condition of it being non-singular is $ad - bc \neq 0$.
- **Invertible Matrix:** The matrix whose inverse matrix can be computed, is called an invertible matrix. The **inverse of a matrix** A is $A^{-1} = \text{Adj } A / |A|$.
- **Conjugate Matrix:** The conjugate matrix of a given matrix is obtained by replacing the corresponding elements of the given matrix, with the complex conjugate of those elements.
- **Transpose Matrix:** The transpose of a matrix A is represented as A^T , and the transpose of a matrix is obtained by changing the rows into columns and columns into rows.
- **Orthogonal Matrix:** If the product of a matrix and its transpose is an identity matrix, then it is called an **orthogonal matrix**. $A \cdot A^T = I$.
- **Hermitian Matrix:** A hermitian matrix is a square matrix, which is equal to its conjugate transpose matrix. The non-diagonal elements of a hermitian matrix are complex conjugates of each other.

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Examples on Unitary Matrix

Example 1: Show that the matrix $A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1i}{\sqrt{2}} & \frac{-1i}{\sqrt{2}} \end{bmatrix}$ is a unitary matrix.

Solution:

The given matrix is $A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1i}{\sqrt{2}} & \frac{-1i}{\sqrt{2}} \end{bmatrix}$

Conjugate of matrix $A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-1i}{\sqrt{2}} & \frac{1i}{\sqrt{2}} \end{bmatrix}$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{-1i}{\sqrt{2}} \end{bmatrix}$$

Practice Questions on Unitary Matrix

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☐ $U^T = U^{-1}$

☐ $U^H = U^{-1}$

☐ $U^H = U$

FAQs on Unitary Matrix

What is the Definition of a Unitary Matrix?

How to Find the Complex Transpose Matrix?

How Do You Know If a Matrix is Unitary Matrix?

What Are the Properties of Unitary Matrix?

is a Unitary Matrix Also a Hermitian Matrix?

What is the Order of a Unitary Matrix?

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Maths Worksheet for Class 7

Maths Worksheet for Class 8

Maths Worksheet for Class 9

Maths Worksheet for Class 10

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Dubai

Australia

France

Germany

Indonesia

Italy

Netherlands

Sri Lanka

Singapore

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Oman

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